MIL-C-43588(GL)

12 July 1968

MILITARY SPECIFICATION

CORD, POLYESTER, WAXED, CORELESS

1. SCOPE

1.1 Scope.- This specification covers the requirements for one size of solid braided polyester waxed cord.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock
CCC-T-191 - Textile Test Methods

MILITARY

MIL-C-3131 - Cordage; Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4020

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:46Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 The requirements specified in 3.5 and 3.6 apply only to cord purchased directly by the Government. All other requirements apply to cord purchased by a supplier as a component for an end item and to cord purchased directly by the Government.

3.2 Materials.- The yarn used in the fabrication of the cord shall be a high tenacity, plied or singles, polyester yarn.

3.3 Construction and physical requirements.- The cord shall be of solid braid construction without core, having 9 strands and conforming to the requirements specified herein.

3.3.1 Diameter.- The diameter of the cord shall be 1/8 inch plus or minus 1/64 inch when tested as specified in 4.2.5.

3.3.2 Length per pound and breaking strength.- The finished waxed cord shall have a minimum of 160 feet per pound and have a minimum breaking strength of 350 pounds when tested as specified in 4.2.5.

3.3.3 Elongation.- The elongation of the cord shall be not more than 25 percent at the 250 pound load when tested as specified in 4.2.5.

3.3.4 Extractable matter (chloroform-soluble material).- The cord shall be waxed and shall have a minimum of 10 percent extractable matter (chloroform-soluble material) when tested as specified in 4.2.5.

3.4 Color.- The color of the finished cord shall be natural. A tan coloration that may be imparted by the wax will be acceptable.

3.5 Put-up.- Unless otherwise specified (see 6.2), the cord shall be furnished, without knots or joining of ends, in continuous length 480-foot spools (reels). The cord shall be wound on the spools (reels) so that each turn and layer is free from entanglement. The ends of the cord shall be served or heat sealed to prevent fraying. A plus tolerance of 10 percent shall be allowed on the length of spool (reel) specified.

3.6 Identification.- Each spool (reel) of cord shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15 point

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:46Z
Check the source to verify that this is the currentversion before use.

paper stock and shall have a minimum tearing resistance of both directions (total) of 850 grams when tested as specified in UU-T-81. When labels are used, the label shall be attached in such a manner as to remain in place and be clearly legible until all cord has been removed. The ticket or label shall be legibly printed, stamped or typed with water insoluble ink. Hand written entries shall be prohibited. The ticket or label shall contain the following information:

Stock Number
Nomenclature
Specification number
Length
Contract number and date
Supplier's name

3.7 Workmanship.- The braided waxed cord shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table I and supplier's certificate of compliance thereto shall be furnished and will be accepted for the requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:46Z
Check the source to verify that this is the current version before use.

TABLE I.- Component tests

Characteristic	Paragraph reference
Material identification	3.2

4.2.2 Examination of the end item for visual defects.- The cord shall be examined by unwinding and the defects listed in table II shall be counted regardless of their proximity to each other. The sample unit for this examination shall be one spool (reel). The lot size for this examination shall be expressed in units of one spool (reel) each. Not less than 100 feet from each sample unit shall be subjected to the visual examination. The acceptable quality level shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

TABLE II.- Visual examination

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Any cut	X	
	Chafed or damaged	X	
	Kink	X	
	Knot	X	
	Unevenly braided resulting in open places, break in continuity of braid, or soft spot	X	
	Broken or loose end or strand	X	
	Loosely braided	X	
Color	Other than specified	X	
	Not uniform		X
Cleanness	Spot or stain 1/		X
Identification ticket or label	Omitted, incorrect, illegible, insecurely attached		X
	Ticket or label not as specified		X

1/ Clearly visible at normal inspection distance (approximately 3 feet).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:46Z
Check the source to verify that this is the current version before use.

4.2.3 Examination for length and winding.- The sample unit for this examination shall be one spool (reel). The inspection level shall be level S-3 and the acceptable quality level shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size shall be 10 and the acceptance number 1. The lot size shall be the number of spools (reels) in the lot. Defects shall be as listed in 4.2.3.1 and 4.2.3.2. Any spool (reel) found to contain one or more defects shall be classified as a defective unit. The length on the spool (see 3.5) shall be determined using the following formula:

W x L = Length per spool (reel) in feet

where: W = net weight of the spool (reel) in 0.1 pounds
L = lot average of the weight-length relationship (length/pound) as determined in 4.2.5

4.2.3.1 Defects with regard to length.- Defects shall be considered to exist if any of the following are determined during inspection:

(a) The length is less than specified or more than 10 percent in excess of the length specified.

(b) Length is less than marked on ticket or label.

4.2.3.2 Defects with regard to winding.- Defects shall be considered to exist if any of the following are determined during inspection:

(a) Improperly or not firmly wound resulting in kinking, knotting, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of cord.

(b) Put-up not as specified i.e. not spools (reels).

(c) Any end not served or heat sealed.

4.2.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging and packing requirements of this specification are complied with. The examination shall be in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the acceptable quality level shall be 2.5 defects per 100 units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:46Z
Check the source to verify that this is the current version before use.

4.2.5 Testing of the end item.- The methods of testing specified in CCC-T-191, whenever applicable and as listed in table IV shall be followed. When the data in the "Number of determinations" or "Results reported as" columns are not specified in the table, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample size shall be in accordance with table III. The lot size shall be expressed in units of one spool (reel). All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, tests shall be made upon material which has reached equilibrium under Standard Conditions as defined in CCC-T-191.

TABLE III.- Sampling for tests

No. of units in lot	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV.- Test methods

Characteristic	Requirement reference	Test method	Number of determinations per sample unit	Results reported as
Construction (solid braid without core)	3.3	Visual	1	Pass or fail
Number of strands	3.3	Visual	1	Pass or fail
Diameter	3.3.1	4.2.5.1	-	-
Length per pound	3.3.2	4.2.5.1	-	-
Breaking strength	3.3.2	4102	5	-

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:46Z
Check the source to verify that this is the current version before use.

TABLE IV.- Test methods (cont'd)

Characteristic	Requirement reference	Test method	Number of determinations per sample unit	Results reported as
Elongation	3.3.3	4102 1/	5	-
Extractable matter (chloroform-soluble material)	3.3.4	2611	-	-

1/ The cord shall be put under a tension load of 4 pounds.

4.2.5.1 Determination of diameter and length per pound.- A specimen of cord shall be put under a tension load of 4 pounds. The diameter of the cord shall be measured by calipers, at the approximate center of the specimen, one minute after application of the tension load and while under tension load. As soon as the diameter measurement has been determined and with the specimen still under load, a minimum length of 3 feet shall be accurately measured and marked on the specimen to the nearest 1/32 inch. After marking, the tension load shall be removed and length measured shall be cut out of the test specimen and accurately weighed to within plus or minus 1.0 percent of its total weight. Five determinations shall be made per sample unit and the results averaged to the nearest 1/64 inch for the diameter and to nearest 0.1 foot for the length per pound.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- Cord shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- Cord shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking.- In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

Source: https://assist.dla.mil -- Downloaded: 202640-21T21:46Z
Check the source to verify that this is the currentversion before use.

6. NOTES

6.1 Intended use.- The cord is intended for use as antenna guys.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Put-up, if other than specified (see 3.5).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Supersession data.- This specification includes requirements of Interim Purchase Description IP/DES S-11-3, dated 25 January 1963, for Cord, Polyester, Waxed, Coreless.

Custodian:

Army - GL

Preparing activity:

Army - GL

Review activities:

Army - MD
DSA - IS

Project No. 4020-A001

User activities:

Army - ME, WC

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:46Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Cord, Polyester, Waxed, Coreless - MIL-C-43588(GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. *IS THE SPECIFICATION RESTRICTIVE?*

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:46Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

☆ U. S. GOVERNMENT PRINTING OFFICE: 1968-341-510/A-629

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:46Z
Check the source to verify that this is the current version before use.